

Research Journal

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September

Sep. 16, 2025 — Tuesday

- Revisited the credentials provided by the HPC facility at IUCAA last year – they are working fine
- Here are the current credentials:

- **VPN**

- * Server: moon.iucaa.in:4433
- * Username: parag.bhattacharya
- * Password: P@^@GBvpn2*2486
- * Domain: LocalDomain

- **ASSC account**

- * Command: ssh -X paragb@192.168.11.34
- * Password: same as Win11

- **Using the installed software**

```
source xraysoftinit
heainit
sas21init
sas21
```

- **Using Python3.10 etc.**

```
module switch python/python310
```

- **LDAP account credentials (at IUCAA)**

- * Username: parag
- * Password: Saturn\$890

Sep. 17, 2025 — Wednesday

- Hopefully I will be able to do some polarimetric study of science data from IXPE, as proposed by Dr. Monmoyuri Baruah yesterday.
- The target for doing some work is 6 – 8 October 2025, during NEMA-XI at RGU, Arunachal Pradesh.
- I have two documents that I need to study:
 1. Bobrikova *et al.* (2024) – sent by Dr. Monmoyuri Baruah
 2. IXPE Quick Start Data Analysis Guide – downloaded from [HEASARC](#)
- To get started with the data analysis, I want to download the IXPE L2 data for two sources:
 1. GX 13+1 – a weakly magnetized neutron star (WMNS) studied by Bobrikova *et al.*
 2. Mrk 501 – a blazar which is provided as an example in the IXPE Quick Start Guide
- The science data are to be downloaded from the [legacy HEASARC browse window](#) as per the following parameters:
 - Object Name or Coordinates: Mrk 501
Mission: IXPE
Observation window: July 2022
 - Object Name or Coordinates: GX 13+1
Mission: IXPE
Observation window: October 2023

Note: The current Xamin browse portal doesn't seem to load

Sep. 21, 2025 — Sunday

- Downloaded the L2 event files for the following dataset:
 1. Mrk 501
 2. GX 13+1
- Commands for file transfer using SCP – from [Contabo Blog](#)
- Commands for Mrk 501
 - Uploaded Mrk 501 .tar file from PC to ASSC server:

```
scp E:\Research\x-ray_polarimetry\data\raw\mrk-501.tar  
paragb@192.168.11.34:/home/paragb/xray-pol/dataset
```

- Navigated to data folder:
`cd /home/paragb/xray-pol/dataset`
- Extracted data from the .tar file
`tar -xvf mrk-501.tar`
- The extracted L2 files are contained in the folder:
`/home/paragb/xray-pol/dataset/01004701/event_l2`
- The data folder has been accordingly renamed as:
`mv -v 01004701/ mrk-501/`
- So the extracted L2 files are now contained in the folder:
`/home/paragb/xray-pol/dataset/mrk-501/event_l2`
- The .gz files are unzipped as:
`gunzip -v *.gz`

Sep. 22, 2025 — Monday

- Downloaded the relevant CALDB files for IXPE to the ASSC server using the commands

```
export CALDB=/home/paragb/caldb/  
cd $CALDB  
wget -v https://heasarc.gsfc.nasa.gov/FTP/caldb/data/  
ixpe/gpd/goodfiles_ixpe_gpd_20250225.tar.gz  
wget -v https://heasarc.gsfc.nasa.gov/FTP/caldb/data/  
ixpe/xrt/goodfiles_ixpe_xrt_20241028.tar.gz  
gunzip -v goodfiles_*.gz  
tar -xvf goodfiles_*.tar  
rm -rv goodfiles_*.tar
```

- Downloaded the data .tar files for Mrk 501 directly to the ASSC server.

Sep. 23, 2025 — Tuesday

- Began download of Heasoft v6.35.2 from remote server using the following command after opening the terminal in the desired folder.

```
wget -v https://heasarc.gsfc.nasa.gov/FTP/  
software/lheasoft/lheasoft6.35.2/heasoft-6.35.2src.tar.gz
```

- Installed MiKTeX using the following commands ([MiKTeX webpage](#)):

```
curl -fsSL https://miktex.org/download/key |  
    sudo gpg --dearmor -o /usr/share/keyrings/miktex.gpg  
echo "deb [signed-by=/usr/share/keyrings/miktex.gpg]  
    https://miktex.org/download/ubuntu jammy universe" |  
    sudo tee /etc/apt/sources.list.d/miktex.list  
sudo apt-get update  
sudo apt-get install miktex  
miktexsetup finish  
initexmf --set-config-value [MPM]AutoInstall=1
```

- Installed TeXstudio using the following commands:

```
sudo add-apt-repository ppa:sunderme/texstudio  
sudo apt update  
sudo apt install texstudio
```

- The .tar data files have been downloaded into the ASSC server at

/home/paragb/xray-pol/dataset/mrk-501/01004701/

- All .gz files have been extracted in the following folders:

auxil event_l1 event_l2 hk

- To view the images on DS9: `ds9 ixpe01004701_det1_evt2_v01.fits &`

- Enhancement of the image:
scale → log
color → heat
- Creation of a source region:
In the menu: e.g. Region → Shape → Circle
edit → region → click + drag to draw the region
Double-click on region to open the region dialog. Center in fk5, radius in arcsec.
- Creation of a background region:
In the menu: e.g. Region → Shape → Annulus
edit → region → click + drag to draw the region
Double-click on region to open the region dialog. Center in fk5, radius in arcsec.

- The region files are created in the L2 event folder. These files contain the following:

- Inside `src.reg`:

```
circle(16:53:51.766,+39:45:44.41,60.0")
```

- Inside `bkg.reg`:

```
circle(16:53:51.766,+39:45:44.41,252.0")
-circle(16:53:51.766,+39:45:44.41,132.0")
```

- Extraction of spectra

- Initializing `xselect`

```
xselect prefix=mrk501
xsel> read event "ixpe01004701_det1_evt2_v01.fits"
```

- Extracting the source spectra (e.g. detector 1):

```
xsel> filter region "src.reg"
xsel> extract SPEC stokes=NEFF
xsel> save spec mrk-501_d01_src_
```

This creates 3 source spectra files, namely

```
mrk-501_d01_src_I.pha
mrk-501_d01_src_Q.pha
mrk-501_d01_src_U.pha
```

- Extracting the background spectra (e.g. detector 1):

```
xsel> clear region
xsel> filter region "bkg.reg"
xsel> extract SPEC stokes=NEFF
xsel> save spec mrk-501_d01_bkg_
```

This creates 3 background spectra files, namely

```
mrk-501_d01_bkg_I.pha
mrk-501_d01_bkg_Q.pha
mrk-501_d01_bkg_U.pha
```

- Finding the appropriate RMF files:

1. Find the observation in the `ixmaster` catalogue using the [HEASARC search form](#)
2. Copy the entry for the object of interest under the `time` column, e.g. 2022-07-09 23:23:24.082 for the Mrk 501 obs. ID 01004701

3. Create a CALDB variable that points to the folder containing the IXPE calibration database as

```
export CALDB=/home/paragb/caldb
```

4. Use the time value in the quzCIF command, for detector unit 1 (DU1) of the gas pixel detector instrument (GPD), as follows:

```
quzCIF IXPE GPD DU1 - MATRIX 2022-07-09 23:23:24.082 -
```

This would give the following output:

```
/home/paragb/caldb/data/ixpe/gpd/cpf/rmf/
  ixpe_d1_20211209_01.rmf 1
/home/paragb/caldb/data/ixpe/gpd/cpf/rmf/
  ixpe_d1_20211209_alpha075_01.rmf 1
```

5. We have to use the file whose name contains the string alpha075 when the events are weighted by either the NEFF or SIMPLE weighting schemes. In our case, we used NEFF while extracting the source and background spectra and so we can create a soft link to the alpha075 file as follows:

```
ln -s /home/paragb/caldb/data/ixpe/gpd/cpf/rmf/
  ixpe_d1_20211209_alpha075_01.rmf mrk-501_d01_rmf.fits
```

This creates a symbolic link file named mrk-501_d01_rmf.fits which can be used as the RMF file in XSPEC later.

- Creating the MRF (for Q and U spectra) and ARF (for I spectrum) files:

1. Sent the following error report to NASA Helpdesk:

Recently I have been introduced to IXPE science data analysis and I have been trying to replicate the results of the walkthrough in section 5 of the IXPE quick start analysis guide.

I have been able to successfully generate the I, Q and U spectra as described in the aforementioned document. However, I am running into an error when I try to create the MRF file for the Q and U spectra.

I have copied the attitude file from the /hk folder into the /event_l2 folder. The input and output of the console are as follows:

```
[paragb@localhost event_l2]$ ixpecalcarf \
> evtfile=ixpe01004701_det1_evt2_v01.fits.gz \
> attfiles=ixpe01004701_det1_att_v01.fits.gz \
> arfout=mrk-501_d01.mrf \
```

```
> specfile=none radius=1.0 weight=1 resptype=mrf
Full path name of a Level 2 attitude file.
[ixpe01004701_det1_att_v01.fits]
attfiles=ixpe01004701_det1_att_v01.fits.gz
[Errno 2] No such file or directory: 'none'
```

The issue persists even when I provide the full path name of the attitude file:

```
[paragb@localhost event_l2]$ ixpecalcarf \
> evtfile=ixpe01004701_det1_evt2_v01.fits.gz \
> attfiles=/home/paragb/xray-pol/dataset/mrk-501/
01004701/event_l2/ixpe01004701_det1_att_v01.fits.gz \
> arfout=mrk-501_d01.mrf \
> specfile=none radius=1.0 weight=1 resptype=mrf
Full path name of a Level 2 attitude file.
[attfiles=ixpe01004701_det1_att_v01.fits.gz]
/home/paragb/xray-pol/dataset/mrk-501/01004701/event_l2/
ixpe01004701_det1_att_v01.fits.gz
[Errno 2] No such file or directory: 'none'
```

I am bamboozled because I have followed the exact same steps as given in the document. I am unable to figure out as to where I have been going wrong.

I would be grateful to receive advice from you in this matter.

Sep. 24, 2025 — Wednesday

- Received the following response from NASA Helpdesk:

Hello,

We were unable to reproduce the error you reported.
To help resolve this issue, please try the following:

1. Execute 'punlearn ixpecalcarf' to clean your parameter file, as we suspect there may be a stale parameter value causing the problem.
2. Please also check your attitude file configuration, as your output shows the attitude file name appearing inconsistently with and without the .gz extension.

Please let us know if this resolves the issue or if you continue to experience problems.

Thanks,
Mihoko Yukita

- Installed Heasoft v6.35.2 on laptop.

Sep. 25, 2025 — Thursday

- Started analysis of data for Mrk 501 locally on my laptop.
- Installed Heasoft 6.35.2 locally on my PC (running on Ubuntu 22.04 LTS).
- Remote access for CALDB set up in the folder:
/home/pararover/Software/astro-soft/caldb
which contains the files alias_config.fits and caldb.config
- The .bashrc file is appended with the following lines:

```
CALDBCONFIG=/home/pararover/Software/astro-soft/caldb/
caldb.config; export CALDBCONFIG
CALDBALIAS=/home/pararover/Software/astro-soft/caldb/
alias_config.fits; export CALDBALIAS
CALDB=https://heasarc.gsfc.nasa.gov/FTP/caldb; export CALDB
```

- Data from HEASARC are downloaded and extracted into the following folder:

```
/home/pararover/Documents/Research/xray-pol/dataset/mrk-501/01004701
```

Sep. 28, 2025 — Sunday

- Downloaded .tar files for IXPE observations GX 13+1, with the Obs. ID. 02006801, in two batches because HEASARC does not allow downloads more than 2 GB.
 1. event_l1/ for ~ 1.6 GB
 2. event_l2/, auxil/ and hk/ for ~ 700 MB
- The above two batches of .tar files need to be extracted and combined together into a single folder named gx13+1/ with the following folder hierarchy:
 - gx13+1/
 - * 02006801

- auxil/
- event_l1/
- event_l2/
- hk/

October

Oct. 01, 2025 — Wednesday

- Found this good paper on polarized X-ray emission from a BH XRB named 4U 1630-47

<https://iopscience.iop.org/article/10.3847/2041-8213/acd77b>

- For their analysis Rawat *et al.* have used IXPEOBSSIM

<https://ixpeobssim.readthedocs.io/en/latest/index.html>

- I intend to explore IXPEOBSSIM further in this direction of research. To this end, I have installed the same on my PC using the command (after running heainit):

```
pip3 install ixpeobssim
```

- IXPEOBSSIM requires the following Python packages to run:

```
numpy    scipy    matplotlib    astropy    regions    skyfield
```

- I have upgrade all of the above packages individually by running the command:

```
pip3 install <<package-name>> --upgrade
```

- Before proceeding, I need to look into the tutorials provided here:

<https://ixpeobssim.readthedocs.io/en/latest/tutorial.html>

- Logs from my tryout of the tutorial:

```
>>> import numpy
>>>
>>> from ixpeobssim.config import file_path_to_model_name
>>> About to create folder /home/pararover/ixpeobssimdata...
>>> Folder succesfully created.
>>> About to create folder /home/pararover/ixpeobssimdata/benchmarks...
>>> Folder succesfully created.
>>> About to create folder /home/pararover/ixpeobssimauxfiles...
>>> Folder succesfully created.
Matplotlib is building the font cache; this may take a moment.
/home/pararover/.local/lib/python3.10/site-packages/matplotlib/
projections/__init__.py:63:
UserWarning: Unable to import Axes3D. This may be due to
multiple versions of Matplotlib being installed
```

```

    (e.g. as a system package and as a pip package). As a result,
    the 3D projection is not available.
    warnings.warn("Unable to import Axes3D.
    This may be due to multiple versions of "
>>> from ixpeobssim.srcmodel.roi import xPointSource, xROIModel
>>> from ixpeobssim.srcmodel.spectrum import power_law
>>> from ixpeobssim.srcmodel.polarization import constant
>>>

```

Oct. 16, 2025 — Thursday

- The solution provided by the IXPE Helpdesk is working.
- Was able to run `ixpecalcarf` to generate MRF files for D01, D02 and D03.
- However, the following warning appeared:

```

pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det2_evt2_v01.fits \
> attfiles=ixpe01004701_det2_att_v01.fits \
> arfout=mrk-501_d02.mrf \
> specfile=None radius=1.0 weight=1 resptype=mrf
/home/pararover/.local/lib/python3.10/site-packages/matplotlib/
projections/_init_.py:63: UserWarning: Unable to import Axes3D.
This may be due to multiple versions of Matplotlib being
installed (e.g. as a system package and as a pip package). As a
result, the 3D projection is not available.
    warnings.warn("Unable to import Axes3D. This may be due to
    multiple versions of "

```

- The `ixpecalcarf` task has been run as follows:

1. For detector #1:

To generate an MRF (for Q and U spectra):

```

pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det1_evt2_v01.fits \
> attfiles=ixpe01004701_det1_att_v01.fits \
> arfout=mrk-501_d01_QU.mrf \
> specfile=None radius=1.0 weight=1 resptype=mrf

```

To generate an ARF (for I spectrum):

```

pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det1_evt2_v01.fits \
> attfiles=ixpe01004701_det1_att_v01.fits \
> arfout=mrk-501_d01_I.arf \
> specfile=none radius=1.0 weight=1 resptype=arf

```

2. For detector #2:

To generate an MRF (for Q and U spectra):

```

pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det2_evt2_v01.fits \
> attfiles=ixpe01004701_det2_att_v01.fits \
> arfout=mrk-501_d02_QU.mrf \
> specfile=none radius=1.0 weight=1 resptype=mrf

```

To generate an ARF (for I spectrum):

```

pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det2_evt2_v01.fits \
> attfiles=ixpe01004701_det2_att_v01.fits \
> arfout=mrk-501_d02_I.arf \
> specfile=none radius=1.0 weight=1 resptype=arf

```

3. For detector #3:

To generate an MRF (for Q and U spectra):

```

pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det3_evt2_v01.fits \
> attfiles=ixpe01004701_det3_att_v01.fits \
> arfout=mrk-501_d03_QU.mrf \
> specfile=none radius=1.0 weight=1 resptype=mrf

```

To generate an ARF (for I spectrum):

```

pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det3_evt2_v01.fits \
> attfiles=ixpe01004701_det3_att_v01.fits \
> arfout=mrk-501_d03_I.arf \
> specfile=none radius=1.0 weight=1 resptype=arf

```

- The list of necessary files for detector #1 are:

mrk-501_d01_src_I.pha

```

mrk-501_d01_src_Q.pha
mrk-501_d01_src_U.pha
mrk-501_d01_bkg_I.pha
mrk-501_d01_bkg_Q.pha
mrk-501_d01_bkg_U.pha
mrk-501_d01.rmfm -> ixpe_d1_20211209_alpha075_01.rmfm
mrk-501_d01_I.arfm
mrk-501_d01_QU.rmfm

```

- The list of necessary files for detector #2 are:

```

mrk-501_d02_src_I.pha
mrk-501_d02_src_Q.pha
mrk-501_d02_src_U.pha
mrk-501_d02_bkg_I.pha
mrk-501_d02_bkg_Q.pha
mrk-501_d02_bkg_U.pha
mrk-501_d02.rmfm -> ixpe_d2_20211209_alpha075_01.rmfm
mrk-501_d02_I.arfm
mrk-501_d02_QU.rmfm

```

- The list of necessary files for detector #3 are:

```

mrk-501_d03_src_I.pha
mrk-501_d03_src_Q.pha
mrk-501_d03_src_U.pha
mrk-501_d03_bkg_I.pha
mrk-501_d03_bkg_Q.pha
mrk-501_d03_bkg_U.pha
mrk-501_d03.rmfm -> ixpe_d2_20211209_alpha075_01.rmfm
mrk-501_d03_I.arfm
mrk-501_d03_QU.rmfm

```

- Commands for spectrum file grouping of detector #1:

1. For *I* spectrum:

```

fthedit mrk-501_d01_src_I.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d01_bkg_I.pha'" longstring=YES

```

```

fthedit mrk-501_d01_src_I.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d01.rmfm'" longstring=YES

```

```
fthedit mrk-501_d01_src_I.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d01_I.arf'" longstring=YES
```

2. For Q spectrum:

```
fthedit mrk-501_d01_src_Q.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d01_bkg_Q.pha'" longstring=YES
```

```
fthedit mrk-501_d01_src_Q.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d01.rmfm'" longstring=YES
```

```
fthedit mrk-501_d01_src_Q.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d01_QU.rmfm'" longstring=YES
```

3. For U spectrum:

```
fthedit mrk-501_d01_src_U.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d01_bkg_U.pha'" longstring=YES
```

```
fthedit mrk-501_d01_src_U.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d01.rmfm'" longstring=YES
```

```
fthedit mrk-501_d01_src_U.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d01_QU.rmfm'" longstring=YES
```

• Commands for spectrum file grouping of detector #2:

1. For I spectrum:

```
fthedit mrk-501_d02_src_I.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d02_bkg_I.pha'" longstring=YES
```

```
fthedit mrk-501_d02_src_I.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d02.rmfm'" longstring=YES
```

```
fthedit mrk-501_d02_src_I.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d02_I.arf'" longstring=YES
```

2. For Q spectrum:

```
fthedit mrk-501_d02_src_Q.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d02_bkg_Q.pha'" longstring=YES
```

```
fthedit mrk-501_d02_src_Q.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d02.rmfmf'" longstring=YES
```

```
fthedit mrk-501_d02_src_Q.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d02_QU.mrf'" longstring=YES
```

3. For *U* spectrum:

```
fthedit mrk-501_d02_src_U.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d02_bkg_U.pha'" longstring=YES
```

```
fthedit mrk-501_d02_src_U.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d02.rmfmf'" longstring=YES
```

```
fthedit mrk-501_d02_src_U.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d02_QU.mrf'" longstring=YES
```

- Commands for spectrum file grouping of detector #3:

1. For *I* spectrum:

```
fthedit mrk-501_d03_src_I.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d03_bkg_I.pha'" longstring=YES
```

```
fthedit mrk-501_d03_src_I.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d03.rmfmf'" longstring=YES
```

```
fthedit mrk-501_d03_src_I.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d03_I.arfmf'" longstring=YES
```

2. For *Q* spectrum:

```
fthedit mrk-501_d03_src_Q.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d03_bkg_Q.pha'" longstring=YES
```

```
fthedit mrk-501_d03_src_Q.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d03.rmfmf'" longstring=YES
```

```
fthedit mrk-501_d03_src_Q.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d03_QU.mrf'" longstring=YES
```

3. For *U* spectrum:

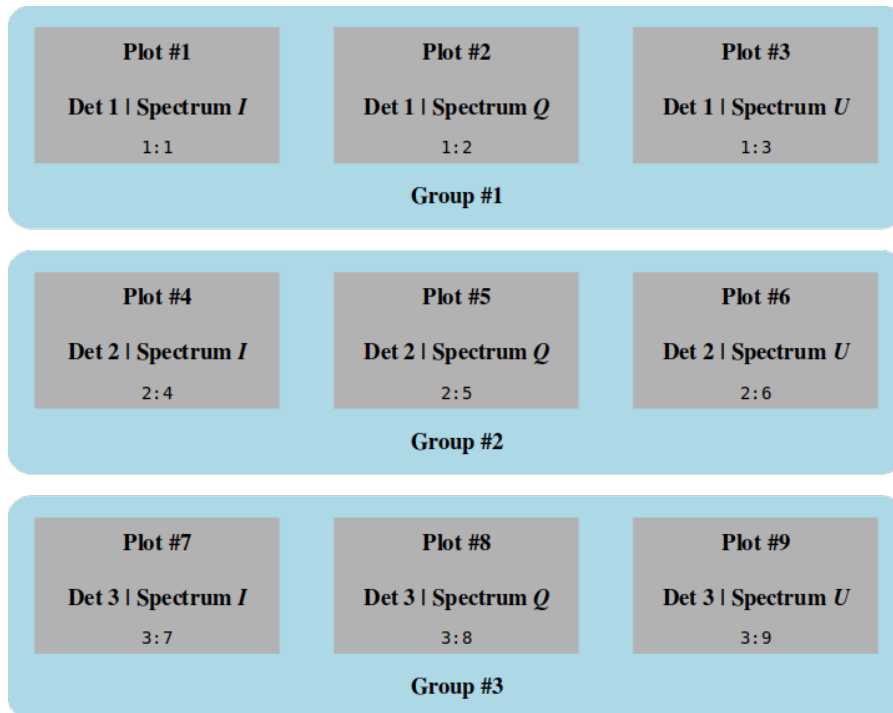
```
fthedit mrk-501_d03_src_U.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d03_bkg_U.pha'" longstring=YES
```

```
fthedit mrk-501_d03_src_U.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d03.rmfm'" longstring=YES
```

```
fthedit mrk-501_d03_src_U.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d03_QU.mrf'" longstring=YES
```

Oct. 17, 2025 — Friday

- Spectral fitting of all data groups to be performed as per the Quick Start Manual.
- Understanding the data and plot groups in XSPEC:



- The above data groups are created in XSPEC as follows:

```
data 1:1 mrk-501_d01_src_I.pha
data 1:2 mrk-501_d01_src_Q.pha
data 1:3 mrk-501_d01_src_U.pha

data 2:4 mrk-501_d02_src_I.pha
```



```
data 2:5 mrk-501_d02_src_Q.pha
data 2:6 mrk-501_d02_src_U.pha
```

```
data 3:7 mrk-501_d03_src_I.pha
data 3:8 mrk-501_d03_src_Q.pha
data 3:9 mrk-501_d03_src_U.pha
```

Oct. 18, 2025 — Saturday

- Steps that were used to analyze the spectra of Mrk 501 in XSPEC are given below:

I. Reading the data:

```
XSPEC12> data 1:1 mrk-501_d01_src_I.pha
XSPEC12> data 1:2 mrk-501_d01_src_Q.pha
XSPEC12> data 1:3 mrk-501_d01_src_U.pha
XSPEC12> data 2:4 mrk-501_d02_src_I.pha
XSPEC12> data 2:5 mrk-501_d02_src_Q.pha
XSPEC12> data 2:6 mrk-501_d02_src_U.pha
XSPEC12> data 3:7 mrk-501_d03_src_I.pha
XSPEC12> data 3:8 mrk-501_d03_src_Q.pha
XSPEC12> data 3:9 mrk-501_d03_src_U.pha
```

II. Filtering the photons within an energy bandwidth:

```
XSPEC12> ignore *:-2.0 8.0-*
```

Here the first * means to apply this filter on all data groups. Then *-2.0 8.0-* means to ignore all photon counts for $E < 2 \text{ keV}$ and $E > 8 \text{ keV}$, i.e. only the $2 \text{ keV} \leq E \leq 8 \text{ keV}$ bandwidth is to be considered.

III. Applying the composite model:

```
XSPEC12> model constant*tbabs*(polconst*powerlaw)
```

Here constant models an energy-independent multiplicative factor, tbabs models the intervening absorption, polconst models a constant polarization at all energies and powerlaw models a simple photon power law.

IV. Parameter #1 is frozen at 1. Parameters 7 and 13 are untied and thawed. The parameter list prior to fitting looks as follows:

```
XSPEC12> show par
```

Parameters defined:

```

=====
Model constant<1>*TBabs<2>(polconst<3>*powerlaw<4>) Source No.: 1
Active/On
Model Model Component Parameter Unit Value
par comp
Data group: 1
1 1 constant factor 1.000000 frozen
2 2 TBabs nH 10^22 1.000000 +/- 0.0
3 3 polconst A 1.000000 +/- 0.0
4 3 polconst psi deg 45.0000 +/- 0.0
5 4 powerlaw PhoIndex 1.000000 +/- 0.0
6 4 powerlaw norm 1.000000 +/- 0.0
Data group: 2
7 1 constant factor 1.000000 +/- 0.0
8 2 TBabs nH 10^22 1.000000 = p2
9 3 polconst A 1.000000 = p3
10 3 polconst psi deg 45.0000 = p4
11 4 powerlaw PhoIndex 1.000000 = p5
12 4 powerlaw norm 1.000000 = p6
Data group: 3
13 1 constant factor 1.000000 +/- 0.0
14 2 TBabs nH 10^22 1.000000 = p2
15 3 polconst A 1.000000 = p3
16 3 polconst psi deg 45.0000 = p4
17 4 powerlaw PhoIndex 1.000000 = p5
18 4 powerlaw norm 1.000000 = p6
=====

```

V. Renormalizing and fitting the model to the spectrum gives the following parameter values:

```

=====
Model constant<1>*TBabs<2>(polconst<3>*powerlaw<4>) Source No.: 1
Active/On
Model Model Component Parameter Unit Value
par comp
Data group: 1
1 1 constant factor 1.000000 frozen
2 2 TBabs nH 10^22 0.541250 +/- 0.117460
3 3 polconst A 7.25453E-02 +/- 1.90183E-02
4 3 polconst psi deg -49.1296 +/- 7.50921
5 4 powerlaw PhoIndex 2.60742 +/- 5.68093E-02
6 4 powerlaw norm 9.63365E-02 +/- 8.13714E-03
Data group: 2

```

7	1	constant	factor		1.00375	+/-	9.94110E-03
8	2	TBabs	nH	10 ²²	0.541250	=	p2
9	3	polconst	A		7.25453E-02	=	p3
10	3	polconst	psi	deg	-49.1296	=	p4
11	4	powerlaw	PhoIndex		2.60742	=	p5
12	4	powerlaw	norm		9.62997E-02	=	p6
Data group: 3							
13	1	constant	factor		0.924722	+/-	1.00195E-02
14	2	TBabs	nH	10 ²²	0.541250	=	p2
15	3	polconst	A		7.25453E-02	=	p3
16	3	polconst	psi	deg	-49.1296	=	p4
17	4	powerlaw	PhoIndex		2.60742	=	p5
18	4	powerlaw	norm		9.63365E-02	=	p6

Fit statistic: Chi-Squared 131.57 using 149 bins,
spectrum 1, group 1.

Chi-Squared	159.05	using 149 bins, spectrum 2, group 1.
Chi-Squared	122.69	using 149 bins, spectrum 3, group 1.
Chi-Squared	130.71	using 149 bins, spectrum 4, group 2.
Chi-Squared	118.58	using 149 bins, spectrum 5, group 2.
Chi-Squared	157.11	using 149 bins, spectrum 6, group 2.
Chi-Squared	60.87	using 149 bins, spectrum 7, group 3.
Chi-Squared	82.38	using 149 bins, spectrum 8, group 3.
Chi-Squared	69.58	using 149 bins, spectrum 9, group 3.
Total fit statistic	1032.53	with 1334 d.o.f.

Test statistic : Chi-Squared 1032.53 using 1341 bins.
Null hypothesis probability of 1.00e+00 with 1334 degrees of freedom

VI. We are interested in finding the errors in the polarization degree A and polarization angle ψ , which are represented by the parameters A and ψ in the `polconst` model component.

- To find the 99% confidence intervals for, say data group 1, we have

```
XSPEC12> error 6.635 3
```

```
XSPEC12> error 6.635 4
```

- To find the 1σ confidence intervals for, say data group 1, we have

```
XSPEC12> error 1.0 3
```

```
XSPEC12> error 1.0 4
```

Note: We cannot compute the errors in the same parameters for the other two data groups (i.e. 2 and 3) because these parameters are tied to their counterparts in data group 1. Any attempt to do so will be flagged as an error by XSPEC as follows:

```
XSPEC12> error 6.635 9
*** Parameter 9 is not a variable model parameter and
    no confidence range will be calculated.
Parameter    Confidence Range (6.635)
XSPEC12> error 6.635 10
*** Parameter 10 is not a variable model parameter and
    no confidence range will be calculated.
Parameter    Confidence Range (6.635)
XSPEC12> error 6.635 15
*** Parameter 15 is not a variable model parameter and
    no confidence range will be calculated.
Parameter    Confidence Range (6.635)
XSPEC12> error 6.635 16
*** Parameter 16 is not a variable model parameter and
    no confidence range will be calculated.
Parameter    Confidence Range (6.635)
```

VII. To visualize the contour plot, we need the following commands:

```
XSPEC12> steppar 3 0.00 0.21 50 4 -90 0 60
XSPEC12> plot contour , , 4 1.386, 4.61 9.21 13.81
```

The steppar command generates a 2D grid of A and ψ values with 50 steps in the range $0 < A < 0.21$ and 60 steps in the range $-90^\circ < \psi < 0^\circ$.

The plot contour command plots contours at 50% (1.386), 90% (4.61), 99% (9.21) and 99.9% (13.81) confidence levels for 2D errors in χ^2 .

- To estimate the MDP_{99} using PIMMS, we need to do the following:

1. Determine the model flux in the $2\text{ keV} \leq E \leq 8\text{ keV}$ bandwidth:

```
XSPEC12> flux 2.0 8.0
Spectrum Number: 1, 2, 3
Data Group Number: 1
Model Flux 0.015886 photons (8.7966e-11 ergs/cm^2/s) range
(2.0000 - 8.0000 keV)

Spectrum Number: 4, 5, 6
Data Group Number: 2
Model Flux 0.015945 photons (8.8297e-11 ergs/cm^2/s) range
```

(2.0000 - 8.0000 keV)

Spectrum Number: 7, 8, 9

Data Group Number: 3

Model Flux 0.01469 photons (8.1345e-11 ergs/cm²/s) range
(2.0000 - 8.0000 keV)

2. Collect the following values from the best-fit parameters so far:

$$n_H = 0.540529 \times 10^{22} \text{ cm}^{-2} \quad (\text{Galactic H column density})$$

$$\Gamma = 2.60715 \quad (\text{photon index from powerlaw})$$

$$\langle F \rangle = \left(\frac{8.7966 + 8.8297 + 8.1345}{3} \right) 10^{-11} = 8.5869 \times 10^{-11} \quad (\text{avg. flux from 3 detectors})$$

3. Enter these values, along with some more, in **WebPIMMS** as shown:



A Mission Count Rate Simulator
Powered by [PIMMS v4.15](#)

Access the [multiple component model interface](#).

Convert From:		Into:	
Flux		IXPE	
Examples of Common FLUX Input/Output Ranges			
Input Energy Range (low-high): 2-8	☒ keV ☐ Angstroms		Units
Output Energy Range (low-high): 2-8	☒ keV ☐ Angstroms		Units

Source Flux / Count Rate: 8.5869e-11		(erg/cm ² /s) (counts/s)
Galactic nH	Redshift	Intrinsic nH
0.540529e22 (cm ⁻²)	none	none (cm ⁻²)

Model of Source:	Model Parameters
☒ Power Law	Photon Index: 2.60715
☐ Black Body	keV:
☐ Therm. Bremss.	kT:
☐ APEC	1.0 Solar Abundance LogT keV

4. **WebPIMMS** computes and displays the predicted count rates and MDP_{99} values using different algorithms, some of which are shown below: We find,

IXPE OPEN Modulation Factor
Modulation Factor = 0.3140 In 10000.0 s, MDP_{99} = 16.13% In 100000.0 s, MDP_{99} = 5.10%
IXPE GRAY TR21D
The Total count rate in the selected band: PIMMS predicts 1.817E-01 cps with IXPE GRAY TR21D
IXPE GRAY ER21D
The Effective count rate in the selected band: PIMMS predicts 1.549E-01 cps with IXPE GRAY ER21D
IXPE GRAY MR21D
The Modulated count rate in the selected band: PIMMS predicts 6.300E-02 cps with IXPE GRAY MR21D
IXPE GRAY Modulation Factor
Modulation Factor = 0.4069 In 10000.0 s, MDP_{99} = 26.80% In 100000.0 s, MDP_{99} = 8.48%

for instance, under ‘IXPE OPEN Modulation Factor’, that for an exposure time of 100 ks, the MDP_{99} is 5.15%.

$$T = 100 \text{ ks} \implies MDP_{99} = 5.10\% = 0.051$$

Now MDP_{99} scales with exposure time T as $MDP_{99} \propto \frac{1}{\sqrt{T}}$. So we have

$$\frac{MDP_{99}(\text{actual})}{MDP_{99}(\text{simulated})} = \sqrt{\frac{T(\text{simulated})}{T(\text{actual})}}$$

Running the show data command gives us the actual exposure time (e.g. for I spectrum by detector #1) to be 97800 s = 97 ks.

Therefore, substituting $T(\text{simulated}) = 100 \text{ ks}$, $T(\text{actual}) = 97 \text{ ks}$ and $MDP_{99}(\text{simulated}) = 5.10\%$ above, we obtain

$$MDP_{99}(\text{actual}) = \sqrt{\frac{100}{97}} (5.10\%) = 5.18\% = 0.0518$$

This implies that, for an *unpolarized source* with the same physical parameters, an IXPE observation will return a value $A > 0.0518$ only 1% (=100%-99%, the 99 is from MDP_{99}) of the time.

- The chain of logic is as follows:

1. Scale PIMMS number because the real observation time was a bit shorter, i.e. $MDP_{99} = 5.18\%$.
2. MDP_{99} means “only 1% chance a zero-polarisation source would look this big or bigger.”
3. Measured value of 7.24% (from best-fit) is above 5.18%, and the measurement’s uncertainty of 1.9% is small (i.e. $5.18/1.9 \approx 3.8$).
4. Therefore the measured polarisation is very unlikely to be a random fluke – it is a highly probable real detection.